ST-2195 Programming for Data Science Report
Coursework Part-2

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1. Introduction

This report enlightens the findings of an analysis from huge datasets via the 2009 ASA Statistical Computing and Graphics Data Expo provided by Harvard Dataverse, which consists of flights within the US. This report focuses on the year 2001 up to 2005, totalling five years. The goal of this report is to identify the best time to reduce delays each year, to show how aircraft age would affect flight delays, and to use the Logistic Regression Model to predict diverted flights in the US.

Using both R programming language in RStudio and Python programming language in Jupyter Notebook, the first step is to import the dataset into R and Python. To prevent errors due to the very large dataset, a crucial step is to remove the diverted and cancelled flights data. This results favourably for the subsequent steps in plotting and visualising the data as it minimises the issue when running the code chunk.

2. Best Times and Days of The Week to Minimise Delays per Year

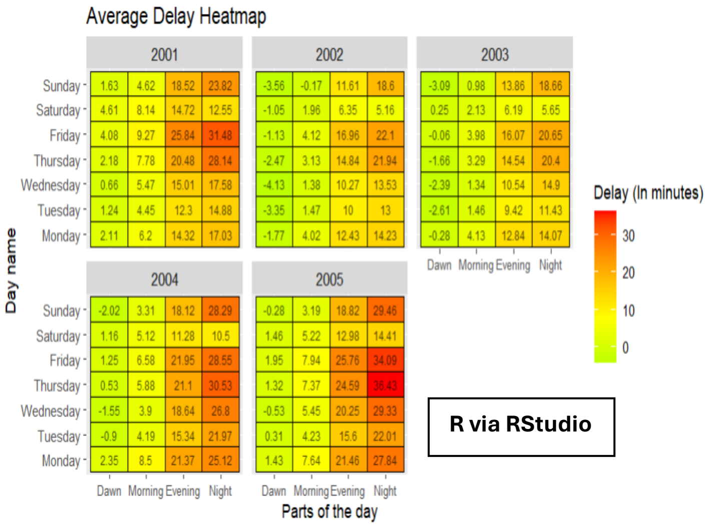
To create a clear observation of the best times and days to minimise flight delays, it is important to create a table that consists of essential elements to observe the average delay in each day and time in each year. To prevent confusion, the table name is disclosed as 'Daily' table in R. The elements used in this table are 'Year', 'DayOfWeek', 'SDepPartOfDay', 'TotalDelays', 'AverageDelay', 'DepDelay', and 'ArrDelay'. In the case for Python, a table called 'flights' is made from copying the table 'flight', which is made by combining each chunk from the SQL Query and converting it into a data frame. Then, the 'TotalDelay' is found by adding 'DepDelay and 'ArrDelay' altogether.

The 'DayOfWeek' element in R and Python is transformed into day names, such as 1 = Monday, 2 = Tuesday, and so on, whereas 'SDepPartOfDay' in R and 'categorized_departure_time' in Python is divided into four times of a day, such as Dawn, Morning, Evening, and Night. With an assumption that all integer data with the value 0 are included in the 'Dawn' category and the value 2400 are included in the 'Night' category, here are the divisions of the times for a day (using a 24-hour clock format)

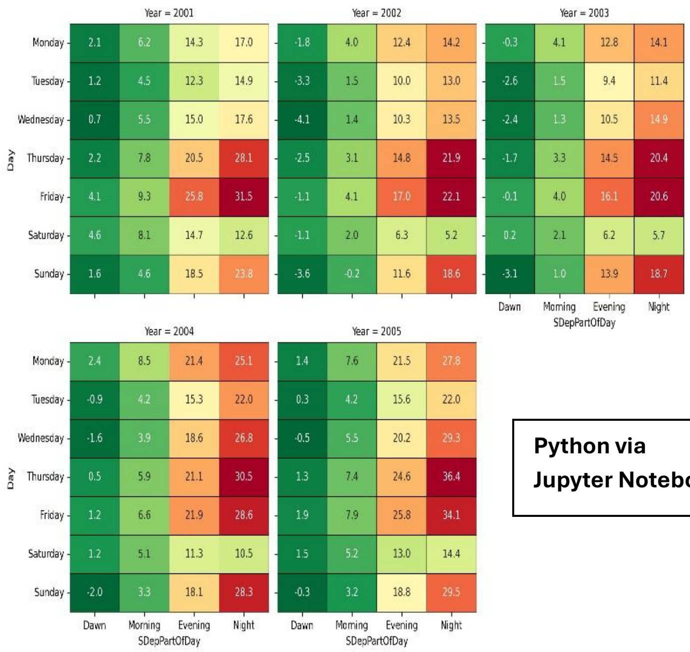
- Dawn (00:00 – 06.00)
- Morning (06:00 – 12:00)
- Evening (12:00 – 18:00)
- Night (18:00 – 24:00)

This assumption is made to prevent any confusion with 00:00 and 24:00 times as some data in the 'CRSDepTime' is not consistent with the time format. In the case for Python, the 'CRSDepTime' will be categorized in 'flights' using the "pd.categorical" function.

After doing some table modifications and calculations, the data can be plotted as a heatmap which shows the average delay of the US flights. The heatmap are as follows:



R via RStudio



Python via Jupyter Notebook

The heatmap consists of parts of the day ("Dawn", "Morning", "Evening", and "Night") as the X-axis and the day names ("Monday", "Tuesday", "Wednesday", "Thursday", "Friday", "Saturday", and "Sunday") as the Y-axis. Inside are numbers that indicate the average delay in minutes. The green coloured boxes indicate the small number in average delay minutes, whereas the red coloured indicates the huge number in average delay minutes. However, there are some cases where the number is negative, meaning that it is not a delay, but rather an early flight in either departure or arrival.

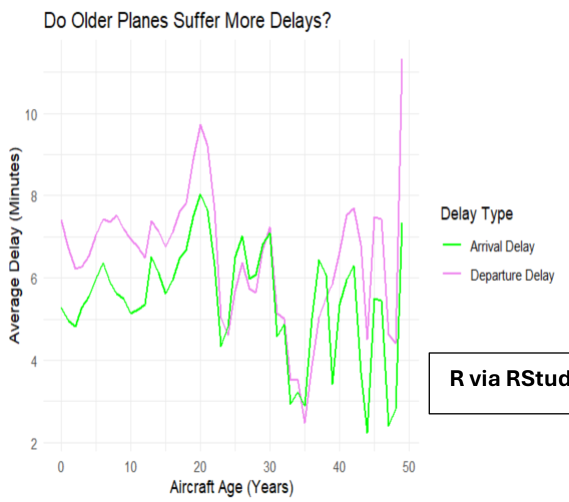
For more clarity, there is a slightly different approach to find these average delays values in R and Python. In R, the average delay is directly found from the AVG function of 'TotalDelays' column in 'Daily' Table. However, the average delay values in Python are found by manually input the total delay divided by the number of flights ('CountOfFlights' from 'flights_counts'). 'Flights_counts' are found by grouping 'flights' by size with included columns such as 'Year', 'Day', and 'SDepPartOfDay'.

According to the heatmap provided, it is shown that most flights have higher chance of getting delayed in the night. This might be due to environment circumstances where there would be more preparation from the aviation team. Another assumption of this issue is that more flights are conducted in the night with the purpose of the passengers will arrive at their destination either in the morning or in the evening. If we filter down by each day, it can be sought that Wednesday dawn time has the lowest average delay.

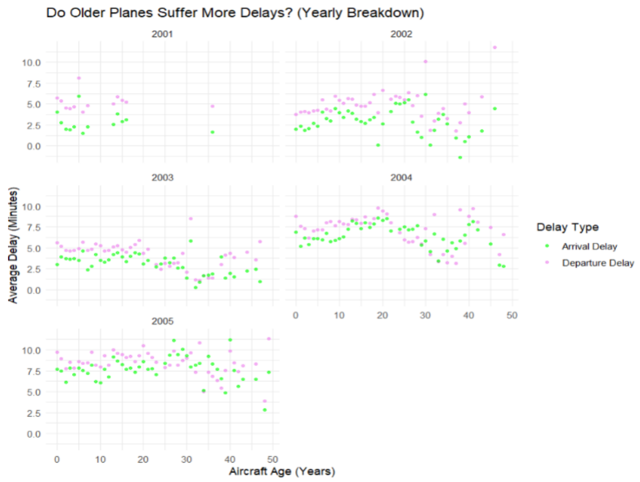
*Therefore, it can be concluded that the best times and days of the week to minimise flight delays each year is on **Wednesday** around **dawn time** with an average of **-1.59 minutes**. A few reasons might be due to the air traffic tends to be quitter in dawn time than other parts of the day. Also, Wednesday being in the middle of the week shows that day is far away from the weekends, where more hectic flight exchanges are usually happening in the weekends.*

3. Do Older Planes Suffer More Delays per Year?

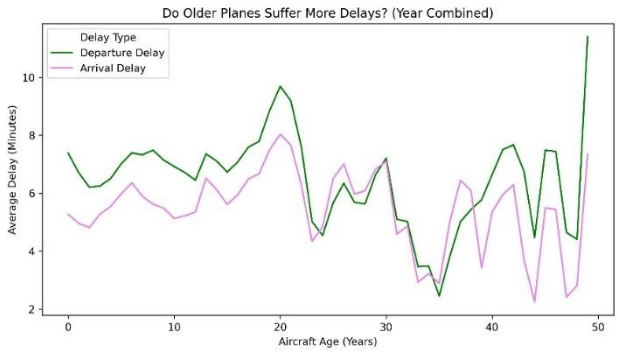
First, merge the 'flight' data with 'planes' data via 'tailnum'. Second, calculate the aircraft age. Third, remove all invalid aircraft age. Complete all these steps only then the average number of plane delays in minutes can be found. In this analysis, there are two viewpoints which can be observed: **line graph and scatter plots**. The difference between both graphs is the base of the year, where the line graph is in combined years and the scatter plot is in an individual year for each.



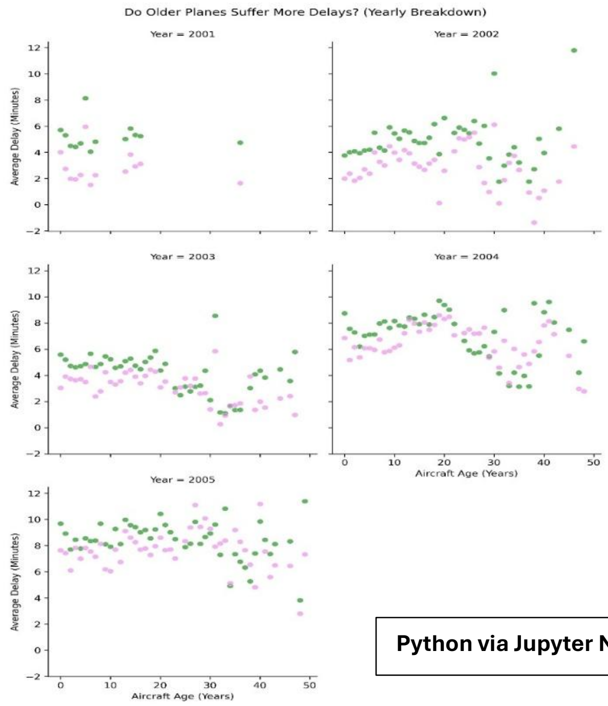
R via RStudio



The average delay which is measured in minutes with the interval of 2.5 minutes for indicator purposes is in the Y-axis while the aircraft age measured in years with the interval of 10 years for indicator purposes is in the X-axis. The colours green and violet represents the delay type, which are arrival delay and departure delay respectively.



The first plot (line graph) shows that **older planes suffer more delays is somewhat true**. This can be seen by the characteristics of the line, where both green and violet lines have the same rising pattern of average delay for the older planes. However, **somewhat true does not mean it is completely true**. Because although middle-aged planes (20-40 years) has a low average delay, it doesn't contradict directly against the statement where older planes suffer more delays.



The second plot (scatter plots) shows that **overall average delay from both arrival and departure delays keeps increasing as the year goes by.** Just by observing the first two years (2001 and 2002) and compare it with the last two years (2004 and 2005) shows the overall increase in average delay no matter the aircraft age is. Relate it with the first plot, the dots with higher point in both X and Y-axis in each year, indicating a longer flight delay is not entirely made by older planes, but rather it is evenly distributed but with older planes are guaranteed to cause longer delays.

Python via Jupyter Notebook

Hence, it is safe to declare that older planes suffer more delays, but it is not absolute as some newer planes caused delays as well. *It is highly possible that older planes suffer more delays due to technical issues as they have older versions of spare parts and might affect the performance of the plane when attempting to taxi. But some newer planes have issues with delays as well due to external factors such as the environment where newer planes are usually assigned in more hectic areas, increasing the chance of being delayed. Also, newer planes are usually assigned in more frequent flights as they tend to be more fuel-efficient, which could raise the probability of delays due to its more recurring nature.*

4. Logistic Regression Model for the Probability of Diverted US Flights

To find the probability of diverted flights in the United States, a previous step of **removing diverted flights must be reverted.** Then, the next step is to prepare the dataset so that they can be used for **training and testing sets of data**. This is crucial for the model to applicate the data patterns from the training datasets into the unseen data in the testing datasets for revealing its accuracy and reliability.

The upcoming step is to choose the relevant variables or attributes that are about to be used for constructing the logistic regression model. These variables include **Day names (R) or days (Python), SDepPartOfDay, CRSElapsedTime, DepDelay, CarrierDelay, WeatherDelay, LateAircraftDelay, and Diverted**, where other than 'Diverted' will work as the predictors. After that, some adjustments such as data type conversion is necessary for the logistic regression to work effectively, such as datasets are converted into factors in R, whereas categorical and numerical are the case in Python.

```
## Create the Model Again with btrain.data Involved
```{r}
logreg_model <- glm(Diverted ~ ., data = btrain.data, family = binomial)
summary(logreg_model)$coef
```

                        Estimate   Std. Error   z value   Pr(>|z|)
(Intercept)   2.199103219 2.723529e-02   80.744616 0.000000e+00
Day_namesMonday   -0.095301000 1.145541e-02   -8.319304 8.848157e-17
Day_namesSaturday   -0.245774964 1.191234e-02   -20.631963 1.417771e-94
Day_namesSunday   0.119746892 1.219316e-02   9.820825 9.159098e-23
Day_namesThursday   0.109260544 1.194771e-02   9.144893 5.968881e-20
Day_namesTuesday   -0.024233218 1.155803e-02   -2.096656 3.602407e-02
Day_namesWednesday -0.030484999 1.161402e-02   -2.624845 8.668847e-03
SDepPartOfDayEvening -0.795101940 2.653252e-02 -29.967069 2.636983e-197
SDepPartOfDayMorning -0.323260222 2.664722e-02 -12.131103 7.226929e-34
SDepPartOfDayNight -0.414774139 2.705071e-02   -15.333207 4.587673e-53
CRSElapsedTime   -0.004542364 4.084912e-05 -111.198560 0.000000e+00
DepDelay   -0.022458747 1.094299e-04 -205.234088 0.000000e+00
CarrierDelay   0.061907130 5.481604e-04 112.936153 0.000000e+00
WeatherDelay   0.038876349 9.588356e-04   40.545376 0.000000e+00
LateAircraftDelay   0.044028833 3.342383e-04 131.728854 0.000000e+00
```

A summary of the logistic regression model is made in R where it shows each of the predictor's effect size estimate, standard error, z-value (standard deviation), and p-value (probability of an observation where it is as extreme or more extreme than the known values given that the null hypothesis is true).

```
        2         4        10        13        15        21
0.8210069 0.8196468 0.8004144 0.8247014 0.7752876 0.8152042
```

These predictions are made with an adjustment in sampling due to overfitting probabilities by up sampling the minor datasets, which in this case is the "Diverted" flights and down sampling the major datasets, which in this case is the "Undiverted" flights. Varying from 0 to 1, where values closer to 1 means that diverted flights are more regular. To conclude whether a flight is diverted or not, a threshold of 0.5 is set for this prediction test. If the shown probability is more than 0.5, the flight will be denoted as 'Diverted' and 'Undiverted' otherwise.

```
Confusion Matrix and Statistics

class.pred   Diverted Undiverted
  Diverted         62       2274
  Undiverted      154     125192

               Accuracy : 0.981
                 95% CI : (0.9802, 0.9817)
    No Information Rate : 0.9983
    P-Value [Acc > NIR] : 1

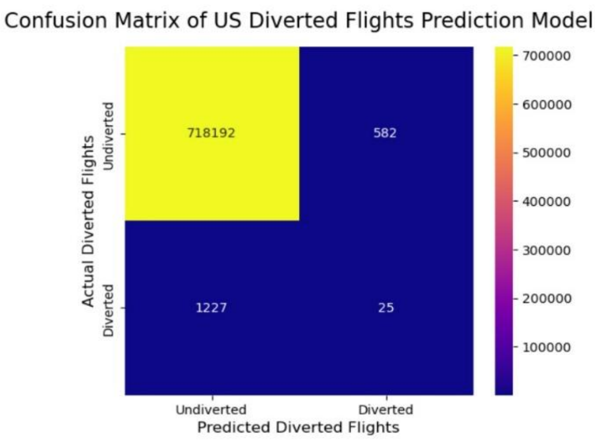
                  Kappa : 0.0456

 Mcnemar's Test P-Value : <2e-16

            Sensitivity : 0.2870370
            Specificity : 0.9821599
         Pos Pred Value : 0.0265411
         Neg Pred Value : 0.9987714
             Prevalence : 0.0016917
         Detection Rate : 0.0004856
   Detection Prevalence : 0.0182955
      Balanced Accuracy : 0.6345985

       'Positive' Class : Diverted
```

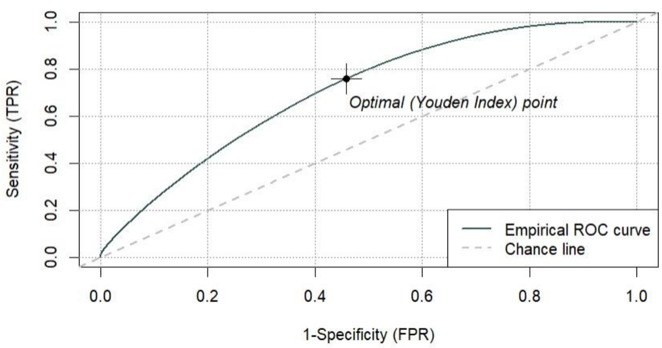
R via RStudio




```
Accuracy: 0.9974875907258904
Precision: 0.04118616144975288
Recall: 0.019968051118210862
```

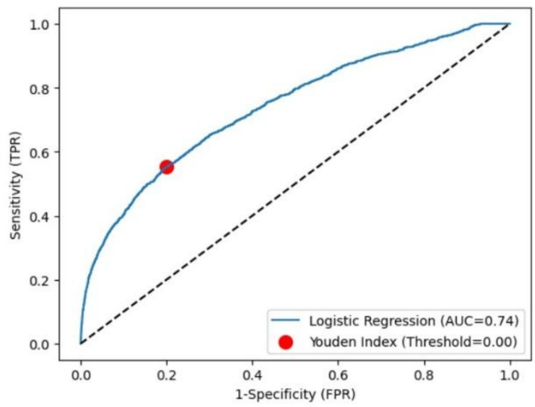
Python via Jupyter Notebook

After computing the predictions from the logistic regression model, a **tester is made to show how accurate and precise this model works** when predicting diverted US flights which is the **Confusion Matrix**. The confusion matrix has a **95% confidence interval of (0.9802, 0.9817) and an accuracy of 98.1% for R and 99.7% for Python.** However, do note that this accuracy has its flaws as NA values are all changed to median and sampling methods are involved here, which could cause some biasness. Also, **sensitivity shows that the model only correctly predicts 28.7% of diverted flights in the US**. This shows that this model is not good for predicting diverted US flights. **The Positive Prediction Value or Recall being only 2.6% (R) and 1.99% (Python) means that the model will 97.4% wrong when it predicts a flight as a "Diverted" flight.** All of these are further being supported by the existing low Kappa value, showing weak predicting capabilities. In conclusion, it may seem accurate at first. However, due to other attributes shown in the confusion matrix which shows the flaws of this model, it can be concluded that **this model is accurate but not suitable for predicting diverted US flights**, especially when the variables in the dataset has very distinct ratio (Diverted vs Undiverted).




```
[1] "ROC score: 0.706071768763866"
```

R via RStudio



Python via Jupyter Notebook

The **ROC Curve is used to further help in the assessment of the logistic regression model by showing the optimal point or known as Youden Index point.** The Youden Index is a measure for ROC curve that is used for seeking optimal thresholds. The optimal Youden Index point means that it is the **best-case scenario of correctly predicting diverted US flights with minimum false predictions**. The Youden Index serves as a measurement of finding the optimal thresholds. By knowing the optimal thresholds, the ROC Curve can reveal when is the best-case scenario of correctly predicting diverted US flights with minimum false predictions.

Both for R and Python shows the AUC or ROC score for this model is more than 0.7 (R = 0.706; Python = 0.74). This score is shown in the area below the ROC curve, where it measures how well this model classifies its results. Therefore, **a score of more than 0.7 shows that the model has a quite remarkable predictive capability.**

5. Conclusion

This report uses a dataset of planes' operation by Harvard Dataverse such as departure and arrival times, many kinds of delays, and diverted flights. By utilising the aforementioned variables, the analysis provides the desirable outcomes for best times and days from a week to minimise flight delays, determining how older planes affect overall delay time, and constructing a logistic regression model for diverted flights in the US.

First, the airport managers and other relevant staff who works in the aircraft schedules may find this report helpful to identify delay factors, thus reducing the chance of delayed flights, or even mitigate delay risks. Minor improvements in the broadcasting section and information systems could be a much of help for many passengers, especially for those who in need, thus may reduce delay potential due to misinformation.

After that, older planes do impact the average delay flight time. However, this only seems to happen in very old planes; roughly 40+ years. It is most likely due to their older components which could not maintain stability for longer flights and needs more time for the aviation team to repair it. But younger planes around the age of 0 to 20 years have also suffered from delays. This is possible because most of these planes are assigned in busier tracks, leading to a higher delay potential.

Last, the logistic regression model shows high accuracy (overall correctness) in predicting flight diversions in the US but not in its precision (positive predictions within classes) for both R and Python, shown by the confusion matrix and ROC curve. Therefore, another approach for modelling strategies is advised for this dataset. Also, not to forget that external factors such as the weather and other disturbances may affect delay potentials and be of use to increase accuracy and precision of diverted US flight predictions.